

Assignment-1.3

“RAM and Storage”

TASK-1 Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

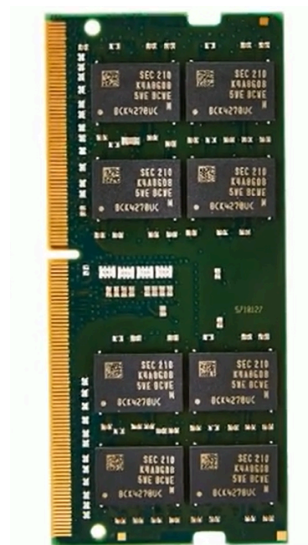
Features	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Speed	800-2133MT/s	1600-3200MT/s	4800-8800MT/s
Voltage	~1.5V	~1.2V	~1.1V
Capacity	Up to 16gb	Up to 64gb	48-128gb+
Use cases	Older PCs/laptops	Many PCs/laptops from the 2010s–early 2020s	Newer PCs/laptops

- ☐ My laptop has DDR4 RAM, and it is not interchangeable because every RAM has different physical notch and electrical characteristics.

TASK-2 Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

- ☐ Reference image of RAM installation in an Acer Aspire Lite laptop , Physical removal was not performed ,because I did not want to risk damaging my laptop.

These are the 2 RAM slots



Memory: 8.0 GB DDR4
Speed: 3200 MT/s
Slots used: 1 of 2
Form factor: SODIMM
Hardware reserved:221 MB

Available 1.0 GB
Cached 981 MB
Committed 10.0/14.3 GB
Paged pool 415 MB
Non-paged pool 389 MB
In use (Compressed) 6.8 GB (52.5 MB)

TASK-3 List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Features	HDD	SSD	NVMeSSD
Speed	Slow	Fast	Very Fast
Main difference	Uses spinning magnetic disks; cheaper and offers large storage	Uses flash memory with no moving parts; faster and more reliable than HDD	Uses PCIe interface, making it much faster than a normal SSD
Use Cases	Google drives/ Photos backup/ Movies	Windows+Microsof t Office	Gaming(Freefire, PUBG)

TASK-4 Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.



- **M.2 NVMe SSD** – This is a solid-state storage drive that uses the NVMe protocol.
- **No moving parts** – It uses flash memory, so it is faster and quieter than an HDD.
- **M.2 connector** – The gold contacts at the top connect the SSD to the motherboard.
- **512 GB capacity** – The label in the picture shows **512GB**, so it can store about 512 GB of data.